

Particle Physics 2 Standard Model: Lecture 6 : New Revolutions in Particle Physics: Standard Model

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Chapter 1

Massive Weak Exchange, Virtual Particles, and Broken Symmetry

The Standard Model group becomes much less mysterious when we read it as a set of allowed motions on the fermion table. Color transformations move vertically among red, green, and blue. Charged weak transformations move horizontally within quark and lepton doublets. Electromagnetic phase transformations supply a third operation. The corresponding gauge bosons are not decorations added after the symmetries have been found: emitting a gauge boson physically performs the same change of state that a generator of the symmetry describes.

That observation sets up the main question of this lecture. A weak vertex has a coupling comparable in size to an electromagnetic vertex, yet beta decay is enormously slower than a typical strong or electromagnetic process. The missing ingredient is propagation. At low momentum, exchanging a massive W costs a factor of order $1/m_W^2$ in the amplitude. We shall develop that statement first in distance space and then in momentum space, follow its consequences through a long discussion of real and virtual particles, and finally turn to the first qualitative model of spontaneous symmetry breaking.

1.1 Reading Gauge Symmetry from the Fermion Table

1.1.1 Color acts vertically

Arrange the six quark flavors in three weak families,

$$(u, d), \quad (c, s), \quad (t, b), \quad (1.1)$$

and give every quark a red, green, or blue color label. The resulting quark table has three color rows and six flavor columns. Color $SU(3)$ mixes the three entries in any one flavor column. For example, one generator may turn a red up quark into a green up quark. In particle language the same operation is accompanied by emission of a gluon carrying the compensating color and anticolor.

The same color transformation must act on every flavor. The gluon that changes a red up quark into a green up quark is the same species of gluon that changes a red top quark into a green top quark. A genuine symmetry operation does not apply to the first generation while ignoring the

others. Thus color motion is vertical and universal:

$$q_r \longleftrightarrow q_g \longleftrightarrow q_b, \quad q \in \{u, d, c, s, t, b\}. \quad (1.2)$$

It changes color but not flavor.

This is the characteristic relation between a gauge group and its gauge bosons. The infinitesimal generators describe transformations among states, and emission or absorption of the corresponding bosons realizes those transformations as physical processes.

1.1.2 Weak transformations act horizontally

The charged weak interaction moves in the other direction. An up-type quark can emit a charged weak boson and become its down-type partner without changing color:

$$u_b \longrightarrow d_b + W^+. \quad (1.3)$$

Exactly the same W^+ occurs in

$$c_g \longrightarrow s_g + W^+, \quad t_r \longrightarrow b_r + W^+. \quad (1.4)$$

The labels r, g, b are spectators. Gluons do not move horizontally across the table, and charged W bosons do not move vertically through color.

Add the electromagnetic phase symmetry $U(1)$, under which every charged field acquires a phase. The first-pass group structure is then

$$SU(3) \times SU(2) \times U(1). \quad (1.5)$$

The cross signs mean that these are independent factors: one acts on color, one on weak doublets, and one by a phase. This is the useful rough picture, but it is not yet the complete electroweak story.¹

1.1.3 The same pattern includes leptons

Below the three color rows place a colorless lepton row:

	u	d	c	s	t	b
Q	$+\frac{2}{3}$	$-\frac{1}{3}$	$+\frac{2}{3}$	$-\frac{1}{3}$	$+\frac{2}{3}$	$-\frac{1}{3}$
lepton partner	ν_e	e^-	ν_μ	μ^-	ν_τ	τ^-
Q	0	-1	0	-1	0	-1

(1.6)

Within every adjacent pair the electric charges differ by one unit. That is why the same charged weak boson can connect u to d , c to s , t to b , and a charged lepton to its neutrino. The W vertex cares about the appropriate quantum numbers and charge difference, not about whether the fermion is a quark or a lepton.

¹Editorial clarification: The exact Standard Model gauge group is conventionally written $SU(3)_c \times SU(2)_L \times U(1)_Y$. The weak $SU(2)_L$ acts only on left-handed doublets, and the final electromagnetic $U(1)_{\text{em}}$ emerges from $SU(2)_L \times U(1)_Y$ after symmetry breaking. The lecture explicitly warns that its first-pass identification of the last two factors is not yet fully consistent; later lectures supply the missing chiral and electroweak structure.

u	d	c	s	t	b
$\frac{2}{3}$	$-\frac{1}{3}$	$\frac{2}{3}$	$-\frac{1}{3}$	$\frac{2}{3}$	$-\frac{1}{3}$
ν_e	e	ν_μ	μ	ν_τ	τ
0	-1				

Figure 1.1: The completed fermion table: three quark-color rows above the colorless lepton row, with the quark charges and three lepton doublets visible.

Lecture frame: 00:18:55.

The muon has the same charge and known intrinsic quantum numbers as the electron, but is about two hundred times heavier. The tau is heavier still. The neutrino masses are very small and were not known in detail in the lecture's discussion. No principle has yet emerged here that explains the observed pattern of family masses.

Question from the audience. Why are leptons, baryons, and mesons called by those names? ^a

Response. The historical Greek roots meant roughly light, heavy, and intermediate. This mass-based naming became misleading. The muon was once called the mu meson because its mass lay between the electron and proton masses, but its interactions show that it is a lepton, much more like an electron than a pion. Today a meson means a quark–antiquark hadron and is a boson; the muon is not a meson.

^aLecture timestamp: 00:19:02.

1.2 The Puzzle of Weakness

1.2.1 A concrete charged-current process

Use neutron beta decay as the working diagram. The valence quark compositions are

$$n = udd, \quad p = uud. \quad (1.7)$$

One down quark makes the transition

$$d \longrightarrow u + W^-, \quad (1.8)$$

and the charged boson produces

$$W^- \longrightarrow e^- + \bar{\nu}_e. \quad (1.9)$$

Together,

$$n \longrightarrow p + e^- + \bar{\nu}_e. \quad (1.10)$$

The other two valence quarks carry through as spectators. At the first vertex the neutron becomes a proton; at the second, the same weak interaction that connects a lepton doublet converts the W^- into the observed electron and electron antineutrino.

1.2.2 Three couplings, three very different impressions

In QED the electron charge e is the vertex amplitude. Squaring it gives the characteristic emission probability, with conventional factors collected in

$$\alpha = \frac{e^2}{4\pi} \simeq \frac{1}{137} \quad (1.11)$$

in natural units. The strong interaction has a coupling g_s , and at a typical hadronic scale its analogue α_s may be of order 0.2. That is much larger than $1/137$, so gluon emission is common when a quark is struck.

Let g_w denote the charged weak coupling. Its square is not fantastically small; at the order-of-magnitude level used here it is comparable to the electromagnetic coupling. Yet a free neutron waits roughly fifteen minutes before decaying, while a strong decay may occur on a hadronic crossing

time. The historical estimate in the lecture uses the older rough figure of twelve minutes, but either number makes the same puzzle unavoidable.²

If g_w is not extraordinarily small, what makes weak processes weak?

The answer cannot lie at the vertices alone. Every Feynman diagram also contains propagators, the factors that describe the intermediate motion from one vertex to another.

1.3 A Massive Propagator in Distance Space

1.3.1 Vertices are not the whole diagram

The beta-decay diagram contains two weak vertices, contributing two powers of g_w , and a W line joining them. Denote the position-space propagator by $\Delta(x - y)$, where x is the emission event and y the absorption or conversion event. It is the amplitude kernel associated with carrying the field disturbance between those events.

The exact relativistic propagator contains more structure than we need. Its important qualitative feature is its dependence on mass. At short spacelike distance it has a power-law singularity, schematically

$$\Delta(d) \sim \frac{1}{d^2}. \quad (1.12)$$

A massless field retains a power-law tail. A massive field, by contrast, is exponentially suppressed once the separation becomes large compared with its Compton scale. The graph therefore bends sharply downward.

1.3.2 Natural units locate the bend

Set

$$c = \hbar = 1. \quad (1.13)$$

Then

$$[m] = [E] = [p] = L^{-1}. \quad (1.14)$$

The familiar relations explain each equality: $E = mc^2$ identifies mass and energy, while px has the dimensions of $\hbar$, so momentum and distance have inverse units. The only length that can be formed from the mass m is therefore

$$d_m \sim \frac{1}{m}. \quad (1.15)$$

Restoring constants gives

$$d_m \sim \frac{\hbar}{mc}, \quad (1.16)$$

the Compton wavelength. A heavier exchanged particle has a shorter effective range.

²Editorial clarification: The modern free-neutron mean lifetime is about 8.8×10^2 s, with ongoing experimental tension between major measurement methods. The lecture's rough lifetime is retained as the pedagogical scale, not as a precision value.

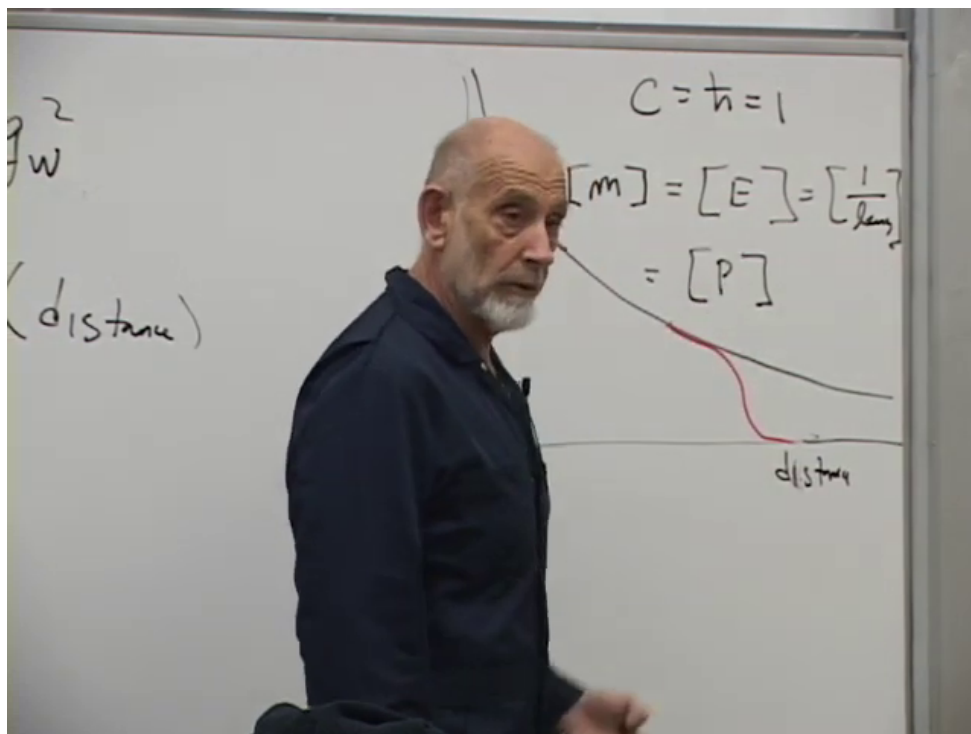


Figure 1.2: Natural-unit dimensions and the qualitative propagator falloff. The black curve continues with a power-law tail; the red massive curve turns down near the inverse-mass scale.

Lecture frame: 00:27:19.

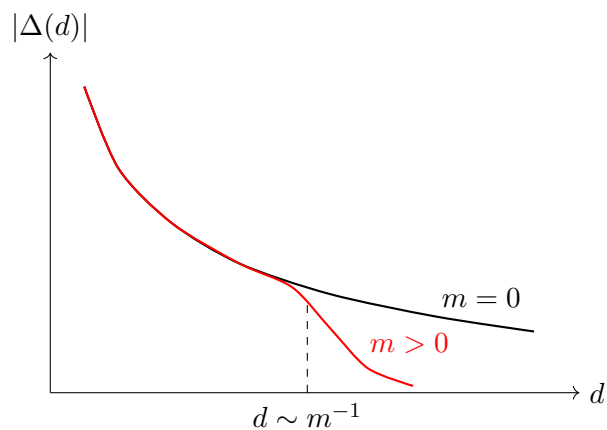


Figure 1.3: A clean schematic of the board argument. It is not an exact formula for a four-dimensional propagator; it isolates the mass-controlled range.

Question from the audience. What is plotted on the vertical axis? ^a

Response. The size of the propagator, denoted by Δ . Its square contributes to a probability after the complete amplitude, external states, and phase-space factors are included. The graph is meant to show that the amplitude for massive exchange over a long distance shuts off rapidly. ^b

^aLecture timestamp: 00:28:30.

^bEditorial clarification: A propagator by itself is not a directly measurable probability amplitude for a particle following a classical path. The lecture's language is a useful range estimate: the propagator is one factor in the full scattering or decay amplitude.

1.4 Fourier Transform and the Fermi Limit

1.4.1 Experiments specify momenta

An experiment usually measures external momenta rather than the two spacetime points at which an internal line begins and ends. In beta decay, the momentum carried by the W is inferred from the difference between the neutron and proton momenta, and it equals the total momentum of the electron and antineutrino. Position-space and momentum-space propagators are related by a Fourier transform.

$$k = p_n - p_p = p_e + p_{\bar{\nu}_e}. \quad (1.17)$$

In the positive-denominator convention used on the board,

$$\tilde{\Delta}_W(k) = \frac{1}{k^2 + m_W^2}, \quad \tilde{\Delta}_\gamma(k) = \frac{1}{k^2}. \quad (1.18)$$

The massless photon propagator becomes large at small momentum transfer. The massive W propagator never exceeds a scale of order $1/m_W^2$. For beta decay,

$$|k^2| \ll m_W^2, \quad (1.19)$$

so the details of k^2 are negligible and

$$\tilde{\Delta}_W(k) \simeq \frac{1}{m_W^2}. \quad (1.20)$$

Question from the audience. Should the relativistic k^2 contain a minus sign between energy and spatial momentum? ^a

Response. Yes. With a mostly-minus Minkowski metric, $k^2 = (k^0)^2 - \mathbf{k}^2$, and the scalar denominator is conventionally $k^2 - m_W^2 + i\epsilon$. The board is using the equivalent positive spacelike or Euclidean magnitude. In low-energy beta decay the transferred four-momentum is tiny compared with m_W , so either notation gives the same $1/m_W^2$ suppression up to an irrelevant sign in the amplitude.

^aLecture timestamp: 00:35:11.

1.4.2 The low-energy amplitude

Two weak vertices and one propagator give

$$\mathcal{A}_{\text{weak}} \sim g_w \tilde{\Delta}_W(k) g_w \quad (1.21)$$

$$\sim \frac{g_w^2}{k^2 + m_W^2} \simeq \frac{g_w^2}{m_W^2}. \quad (1.22)$$

Squaring the amplitude produces the characteristic suppression

$$|\mathcal{A}_{\text{weak}}|^2 \sim \frac{g_w^4}{m_W^4}, \quad (1.23)$$

before the kinematic and phase-space factors appropriate to the particular decay are included. The weak interaction is weak at nuclear energies not because its elementary gauge coupling is absurdly small, but because its mediator is heavy.

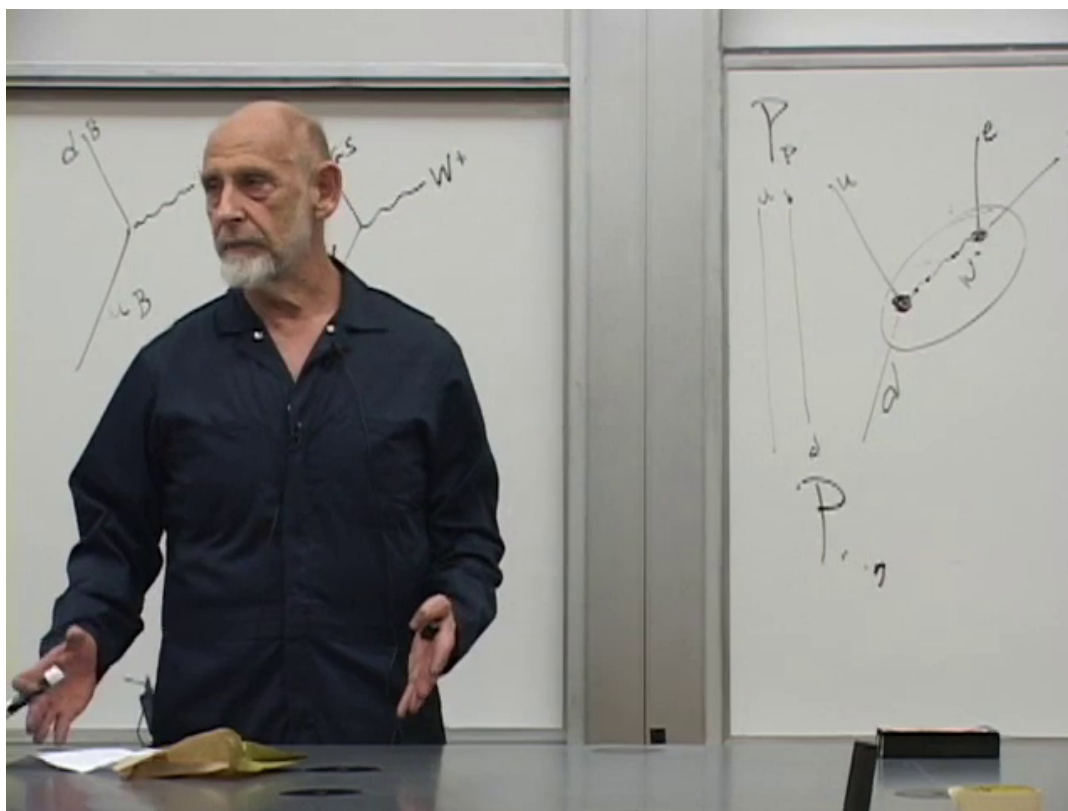


Figure 1.4: The neutron–proton weak process and the momentum transfer carried by the internal line, as discussed while fixing the propagator sign.

Lecture frame: 00:35:40.

Numerically m_W is roughly one hundred proton masses at the accuracy of this lecture. A factor of m_W^{-2} in the amplitude becomes m_W^{-4} in a rate, so even a coupling of ordinary gauge-theory size produces a very long lifetime.

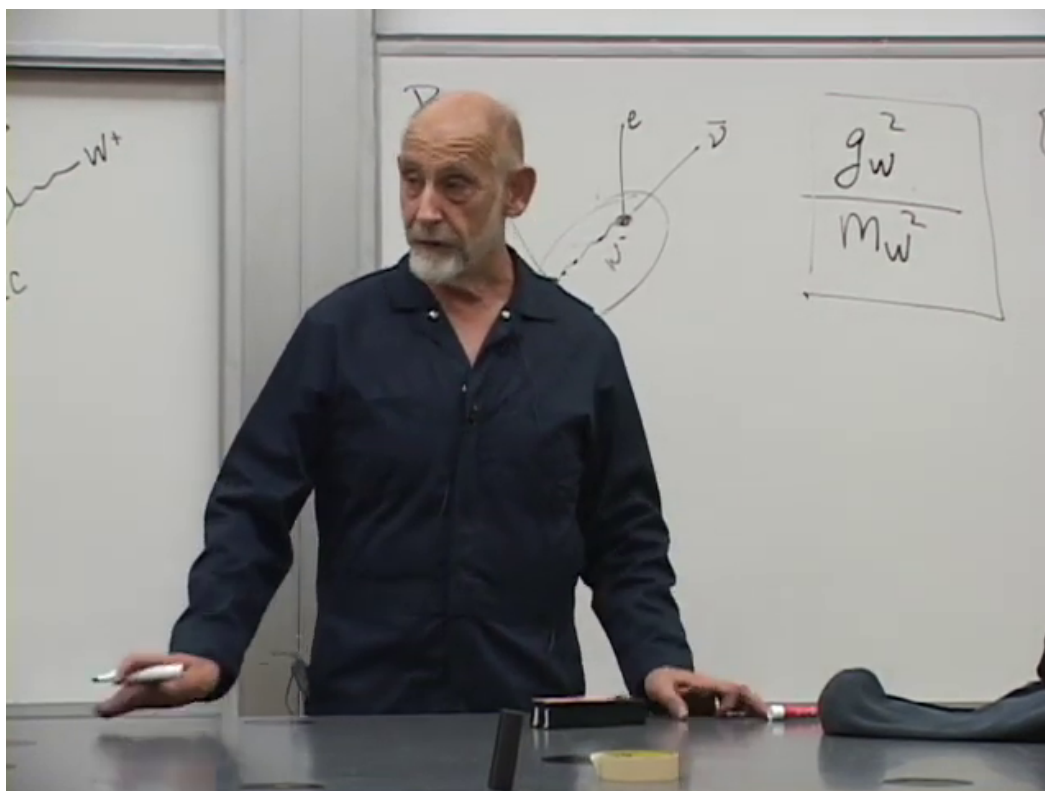


Figure 1.5: The beta-decay topology beside the boxed low-energy estimate g_w^2/m_W^2 .
Lecture frame: 00:37:25.

Question from the audience. Are we talking about a suppression of one in a million or one in a billion, and how was the W mass originally known? ^a

Response. The lecture’s rough counting points toward the billion-scale smallness of a low-energy weak probability relative to an unsuppressed hadronic process. Historically, low-energy experiments did not separately measure g_w and m_W . They measured the small combination g_w^2/m_W^2 , called the Fermi constant. Once the gauge coupling was tied to the electroweak theory, that combination predicted a very heavy mediator; later accelerators produced the W directly and measured its mass.

^aLecture timestamp: 00:38:11.

At momenta far below m_W , every experiment sees nearly the same constant:

$$G_F \sim \frac{g_w^2}{m_W^2}. \quad (1.24)$$

This is why Fermi could describe beta decay in the 1930s as a pointlike four-fermion interaction, long before the mediator was observed. As collision energies increased, experiments became sensitive to the momentum dependence hidden in the propagator, confirming that the effective constant was the low-energy shadow of massive exchange.³

Historically, the idea that weak interactions might be mediated by a heavy particle emerged in the mid-twentieth century; the lecture mentions Julian Schwinger among the theorists who explored that possibility and notes that the letter W probably stood for weak. Low-energy data already implied that such a mediator would have to be extremely heavy on the nuclear scale, long before accelerators had enough energy to make it as a real particle.

1.4.3 Three independent controls on a rate

A classroom question about nuclear binding separates three logically different controls on a transition:

1. vertex coupling constants;
2. propagators, including masses and momentum dependence;
3. available initial-to-final energy and phase space.

Binding can close a decay channel because the final state would have greater energy than the initial state. That is a kinematic prohibition, not a further reduction of g_w or a change in m_W . When a channel remains open but has little available energy, its phase space can still be strongly suppressed.

³Editorial clarification: With standard electroweak normalization at tree level,

$$\frac{G_F}{\sqrt{2}} = \frac{g^2}{8m_W^2}.$$

The lecture deliberately suppresses numerical and spinor factors and keeps only the scaling $G_F \sim g_w^2/m_W^2$. The W and Z were directly discovered at CERN in 1983; the lecture’s passing mid-to-late-1970s dating is too early for the direct observation.

Question from the audience. Does the field or binding of a nearby proton suppress beta decay through the same propagator mechanism? ^a

Response. Not in the example being discussed. Binding matters because it changes the energies of the initial and final nuclear states. A process cannot occur if the final state is energetically inaccessible. Couplings, propagators, and available energy are three separate ingredients.

^aLecture timestamp: 00:41:13.

Question from the audience. Are the coupling constants really constant? ^a

Response. Only at a fixed scale. Quantum corrections make effective couplings run with momentum or distance. The present argument treats each coupling at the scale relevant to the process; scale dependence will be developed later.

^aLecture timestamp: 00:42:55.

1.5 Virtual Exchange without Abandoning Conservation Laws

1.5.1 Why the heavy line can appear

Ordinary neutron decay does not contain enough energy to produce a free, on-shell W . The W in the diagram is internal and off shell. The lecture introduces this fact with the traditional energy–time estimate

$$\Delta E \Delta t \lesssim \hbar : \quad (1.25)$$

an intermediate configuration whose classical energy appears wrong by ΔE can contribute only over a time of order $\hbar/\Delta E$.

The tunneling analogy makes the intended picture vivid. Put an alpha particle in a nuclear potential well. Its energy lies below the top of the barrier, so classically it cannot escape. Quantum mechanically the wavefunction extends through the barrier, and the particle can emerge at a point on the far side with the same total energy it had initially. If we try to describe the interior of the barrier as an ordinary classical trajectory, the potential energy seems too large and the kinetic energy would have to be negative. That failed classical picture is analogous to trying to regard the internal W as a real particle.

In the beta-decay picture the neutron may fluctuate into a proton plus an internal W^- , but the heavy line must be removed quickly. It may be reabsorbed, restoring the neutron, or it may terminate at the second vertex and produce the much lighter electron and antineutrino. The full initial and final energies agree.⁴

1.5.2 Real W Bosons

A sufficiently energetic accelerator can produce a real W . It then has its own proper lifetime, unrelated to a putative uncertainty time for the virtual line. A boosted real W lives longer in the

⁴Editorial clarification: Energy and momentum are conserved at every Feynman vertex. Virtual particles do not literally borrow energy and repay it later. The precise statement is that an internal line is integrated over four-momenta that need not satisfy the real-particle mass-shell relation $k^2 = m_W^2$. The uncertainty-principle and tunneling language is retained because it is the lecture’s explanatory route, but the off-shell statement is the modern QFT formulation.

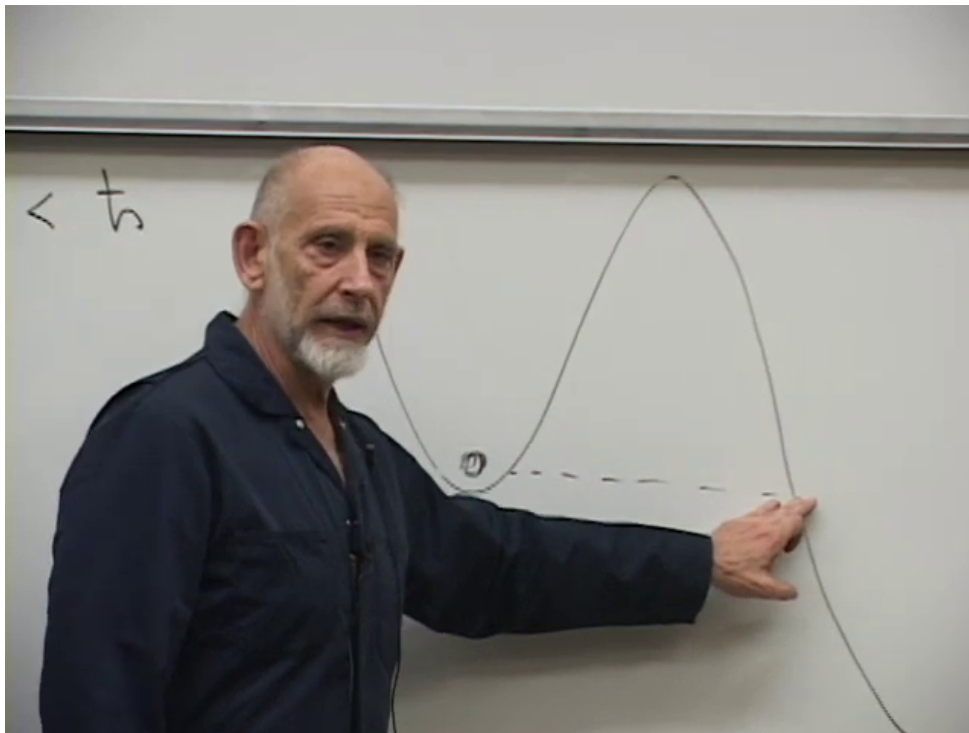


Figure 1.6: The tunneling analogy: the initial point and the marked point on the far slope have equal total energy, while the intervening barrier is classically forbidden.

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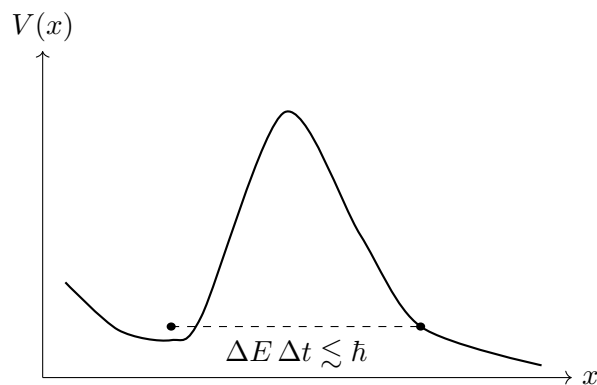


Figure 1.7: A readable reconstruction of the barrier and equal-energy points.

laboratory by ordinary relativistic time dilation. In a two-body leptonic decay of a W at rest,

$$W^- \longrightarrow e^- + \bar{\nu}_e, \quad (1.26)$$

the electron and antineutrino carry large, nearly back-to-back kinetic energies, each of order $m_W/2$. The rest energy has not disappeared; it has become final-state energy. Muon and tau channels, as well as quark channels, are also available.

Question from the audience. How long does a real W live, and can its lifetime be computed from the same coupling? ^a

Response. Yes. One calculates each allowed decay amplitude, squares it, integrates over final-state phase space, and sums the partial widths. The inverse total width gives the lifetime. The transcript's spoken numerical estimate is garbled, but the qualitative point is that the proper lifetime is extremely short and a moving W is time-dilated in the usual way. ^b

^aLecture timestamp: 00:49:49.

^bEditorial clarification: Using the measured width $\Gamma_W \simeq 2.1 \text{ GeV}$ gives $\tau_W = \hbar/\Gamma_W \simeq 3.2 \times 10^{-25} \text{ s}$. This modern number resolves the corrupted numerical exchange without attributing it to the lecture.

Question from the audience. How does a squared amplitude become a rate and hence a lifetime? ^a

Response. The squared matrix element is combined with the density of available final states, conservation delta functions, and normalization factors to obtain a probability per unit time, the decay width Γ . The survival probability is $e^{-\Gamma t}$ in units with $\hbar = 1$, so $\tau = 1/\Gamma$, or $\tau = \hbar/\Gamma$ with units restored. A small weak width therefore means a long weak lifetime.

^aLecture timestamp: 00:50:52.

Question from the audience. Can one assign a velocity to a particle tunneling across the barrier? ^a

Response. One can divide an inferred distance by an inferred transit time, but that quantity should not be interpreted as the velocity of a classical particle or a signal. Attempting to measure a trajectory inside the barrier changes the state and destroys the original tunneling experiment.

^aLecture timestamp: 00:52:42.

1.6 What Does It Mean to Observe a Virtual Particle?

1.6.1 External particles and internal lines

Real particles are the incoming and outgoing asymptotic states that can be prepared or detected far from the interaction region. Virtual particles are internal lines in a perturbative representation of the amplitude. They are not additional hidden detector tracks.

Question from the audience. Is there a clear definition of the difference between a real and a virtual particle? ^a

Response. A real particle is an external on-shell state entering or leaving the interaction region. A virtual particle is an internal off-shell line in a Feynman diagram. The latter is part of the mathematical expansion, not an independently observed object.

^aLecture timestamp: 00:55:05.

To resolve a process lasting only Δt , a probe needs frequencies of order $1/\Delta t$ and hence energy of order $\hbar/\Delta t$. A photon sharp enough to resolve an alpha particle beneath a high barrier can supply the energy needed to lift it into the barrier region. Likewise, a probe energetic enough to resolve the short-lived internal W can create a real W . The attempted observation turns the original process into a different experiment.

The lecture phrases this as a recurring quantum-mechanical frustration: the intervention required to check the forbidden intermediate picture supplies the very energy that makes the apparently forbidden configuration real.

1.6.2 Fluctuations of a proton

In perturbation theory a proton state contains amplitudes for configurations such as

$$p \longleftrightarrow n + W^+, \quad (1.27)$$

and a neutron similarly has

$$n \longleftrightarrow p + W^-. \quad (1.28)$$

If no other interaction intervenes, the internal boson is reabsorbed. A hard probe can instead leave a W among the outgoing particles. One may call that act knocking out a fluctuation or creating a W . The language differs; the measurable statement is the same final-state amplitude.

Question from the audience. When a hard probe produces an outgoing W , did it knock out a particle that was already there or create a new one? ^a

Response. Either picture can organize the mathematics. One time-ordered description says that the proton fluctuated to $n + W^+$ and the probe removed the neutron before reabsorption; another says the collision created the final W . What is invariant is the calculated probability for the observed final state, not a unique classical story about what existed before measurement.

^aLecture timestamp: 01:01:20.

Increasing the collision energy does not expose a fixed inventory of objects that had been hidden inside the target. It opens many possible final states: one W , several bosons, quark jets, hadrons, and other particles allowed by the conserved quantum numbers. Striking even an elementary field hard enough can produce protons and many other composite objects. What comes out need not have been present beforehand as a constituent. The lecture pushes the point to comic extremes: if energy and quantum numbers allow it, even tables, chairs, or a hydrogen atom are not logically excluded as fantastically unlikely final arrangements. The joke carries a serious warning against defining elementarity by a naive inventory of collision debris.

1.6.3 Vacuum fluctuations and temperature

A fluctuation attached to a proton is not the same as a vacuum diagram. The first is a component of an interacting proton state; a vacuum fluctuation has no external proton line. In either case, a sufficiently energetic probe may produce real particles corresponding to an allowed final state. Two energetic photons, for example, can produce an electron–positron pair when the center-of-mass energy exceeds threshold.

Question from the audience. Are these reactions dependent on temperature? ^a

Response. The elementary amplitude does not require a thermal bath. Temperature matters statistically: a high-temperature system contains many particles with high kinetic energy, so threshold-crossing collisions become common. Heating a system containing a bound alpha particle may also supply enough energy for over-the-barrier escape in addition to quantum tunneling. A few isolated energetic particles constitute a beam; many thermally distributed energetic particles constitute a hot bath.

^aLecture timestamp: 01:05:17.

Question from the audience. Is the W emitted in a proton fluctuation a fluctuation of the vacuum? ^a

Response. No. The internal W in $p \leftrightarrow n + W^+$ is a fluctuation of the interacting proton state. A closed virtual process with no external proton is a vacuum fluctuation. They obey the same field-theory rules but belong to different amplitudes.

^aLecture timestamp: 01:06:56.

Question from the audience. Why is the antineutrino required at the leptonic W vertex? ^a

Response. It supplies the conserved lepton quantum numbers and is part of the basic charged-current vertex connecting e , ν_e , and W . The bar distinguishes the outgoing electron antineutrino from a neutrino.

^aLecture timestamp: 01:08:51.

1.7 How Far Can Energy Conversion Be Pushed?

The class next drives the rule that energy can create particles to an intentionally absurd limit. Could a coffee bean become a chocolate cake? There is no practical process of that kind, but relativistic quantum theory does not impose a rule that final matter must resemble the initial object. It requires enough energy, the correct conserved charges, and a nonzero amplitude. A kilogram has an energy of order

$$Mc^2 \sim 9 \times 10^{16} \text{ J}, \quad (1.29)$$

so any microscopic collision that assembled a particular kilogram-scale object would be beyond fantastically improbable.

The Planck mass,

$$m_P = \sqrt{\frac{\hbar c}{G}} \simeq 2.2 \times 10^{-5} \text{ g}, \quad (1.30)$$

is not a maximum possible mass. It is the scale at which gravity becomes strong for a quantum localized to its Compton wavelength. A kilogram contains roughly 5×10^7 Planck masses.

Question from the audience. If collisions exceed the Planck scale, can arbitrarily heavy objects such as chairs or planets emerge? ^a

Response. At trans-Planckian center-of-mass energy concentrated in a small region, gravitational collapse is the overwhelmingly natural outcome: a black hole. A highly organized macroscopic object would be an extraordinarily rare fluctuation. The lecture invokes time reversal: because a chair can fall into a black hole, the reverse amplitude is not exactly zero. Its probability is suppressed by an enormous entropy factor, so possibility must not be confused with practical likelihood.

^aLecture timestamp: 01:10:37.

This excursion reinforces the earlier warning. Quantum field theory predicts amplitudes between specified initial and final states. A picturesque claim that every product was literally contained in the target before impact is not part of the prediction.

1.8 Bosons, Their Interactions, and Their Statistics

1.8.1 Photons can scatter

Being a boson does not mean being noninteracting. Boson or fermion specifies exchange statistics; interaction vertices are separate dynamical facts. Classical light beams normally pass through one another because QED has no tree-level four-photon vertex and low-energy photon–photon scattering is extremely small. Nevertheless, two photons can interact through a virtual electron–positron loop:

$$\gamma + \gamma \longrightarrow (e^- e^+)_{\text{virtual}} \longrightarrow \gamma + \gamma. \quad (1.31)$$

When the center-of-mass energy is high enough, they may instead create a real pair,

$$\gamma + \gamma \longrightarrow e^- + e^+. \quad (1.32)$$

Optical photons have energies of a few electron volts, while the pair threshold is of order $2m_e c^2 \approx 1.02 \text{ MeV}$. The virtual process is therefore heavily suppressed at ordinary optical energies, which is why crossing light beams appear not to notice each other.

Question from the audience. Do virtual W bosons interact with other bosons simply because they are bosons? ^a

Response. They can interact, but not merely because they obey Bose statistics. Gauge vertices determine the dynamical interactions. Statistics instead determines how multiparticle amplitudes behave when identical particles are exchanged. Photons interact weakly through charged virtual particles, and the electroweak theory also contains direct non-Abelian W -boson self-interactions.

^aLecture timestamp: 01:12:47.

1.8.2 Bose enhancement is not a force

Fermion versus boson statistics becomes conspicuous when several identical particles occupy nearby quantum states. Bosons favor multiple occupation; fermions exclude identical occupation. In a typical high-energy collision, particles fly into well-separated momentum states, so this statistical effect may be less obvious than it is in a laser or condensate.

In principle, W bosons possess the same Bose enhancement. The lecture jokes that one could imagine a system that lases W bosons. Their enormous mass and ultrashort lifetime make such a device fantastically impractical, but the statistics is indeed bosonic.

Question from the audience. What color is a proton or neutron? ^a

Response. A physical hadron is a color singlet, often called colorless or white. The shorthand says that a baryon contains one red, one green, and one blue quark. More precisely, its color wavefunction is the antisymmetric superposition

$$|\text{singlet}\rangle = \frac{1}{\sqrt{6}}\epsilon_{abc}|q^a q^b q^c\rangle,$$

not a classical assignment of a permanent color to each named quark.

^aLecture timestamp: 01:18:41.

Question from the audience. Will Bose condensation enter this course? ^a

Response. Yes, through the Higgs phenomenon. A nonzero Higgs field throughout the vacuum is analogous in important ways to a bosonic condensate. The analogy will be developed after spontaneous symmetry breaking has been introduced. ^b

^aLecture timestamp: 01:19:25.

^bEditorial clarification: The Higgs vacuum expectation value is not literally an ordinary gas of on-shell Higgs particles occupying one state. Condensation is an illuminating analogy for a coherent bosonic field and a chosen vacuum, not a complete microscopic identification.

1.9 The Missing Neutral Weak Boson

Return now to the group table. If the horizontal weak symmetry is $SU(2)$, then it has

$$\dim SU(2) = 3 \tag{1.33}$$

generators and therefore three gauge fields. The charged combinations W^+ and W^- account for only two. A third, neutral field is required. It is closely related to the observed Z boson, but it is not simply the photon. The precise neutral combinations involve both the third $SU(2)_L$ field and the $U(1)_Y$ field. That mixing, together with the correct definition of electric charge, is deliberately postponed to the next lecture.

1.10 Spontaneous Symmetry Breaking

1.10.1 A symmetric choice can have an unstable indecision

Imagine a circular dinner table at which each place setting has a utensil to the left and one to the right. Before anyone moves, left and right are equivalent. Once one guest reaches to the right, every neighboring choice is forced and the whole table settles into the right-handed arrangement. The rules were symmetric, but the realized configuration selected one of two possibilities. The story is only an analogy, yet it isolates the combination we need: symmetry, multiple equivalent outcomes, and an instability that lets an arbitrarily small disturbance select one outcome globally.

1.10.2 The heads–tails lattice

For a cleaner model, put a two-state variable $s_i = \pm 1$ at every site of a lattice. Call +1 heads and –1 tails. Let neighboring sites prefer to align:

$$H = -J \sum_{\langle ij \rangle} s_i s_j, \quad J > 0. \quad (1.34)$$

Parallel neighbors contribute lower energy than antiparallel neighbors. There are therefore two degenerate ground states,

$$s_i = +1 \quad \text{for every } i, \quad \text{or} \quad s_i = -1 \quad \text{for every } i. \quad (1.35)$$

The Hamiltonian is invariant under the global flip $s_i \rightarrow -s_i$, but either chosen ground state is not.

Now fix one very distant boundary spin to heads. In an ordinary short-range system we might expect one far-away degree of freedom to have no influence here. In the ordered phase, however, that boundary condition selects the all-heads vacuum. Once selected, a single tail in the sea of heads costs energy because it disagrees with its neighbors. Local excitations are defined relative to the chosen ground state even though the underlying equations retain the heads–tails symmetry.

1.10.3 Explicit breaking and domain walls

Explicit breaking would add a local bias,

$$H_{\text{explicit}} = -J \sum_{\langle ij \rangle} s_i s_j - h \sum_i s_i, \quad h \neq 0. \quad (1.36)$$

Then heads and tails do not have equal energy even at one isolated site, and there is only one preferred vacuum. In spontaneous breaking $h = 0$: the local rules are symmetric and the two vacua remain degenerate, but the system occupies one of them.

Insist instead that the far-left boundary is all heads and the far-right boundary all tails. The lowest-energy configuration must interpolate between the two vacua. The interface is a domain wall.

Question from the audience. What distinguishes spontaneous symmetry breaking from ordinary explicit symmetry breaking? ^a

Response. With explicit breaking the local Hamiltonian already assigns different energies to heads and tails, so there is no pair of symmetry-related ground states. With spontaneous breaking the equations retain the symmetry, the vacua are degenerate, and boundary conditions or an arbitrarily small disturbance select one. The possibility of a domain wall interpolating between distinct vacua is a diagnostic of the latter situation.

^aLecture timestamp: 01:28:36.

The model is deliberately primitive, but it contains the essential logic that will be transferred to the Higgs field: a symmetric theory, an unstable or degenerate vacuum structure, a selected ground state, and physical consequences determined by that choice.

1.11 Where the Lecture Leaves Us

The fermion table turns the Standard Model group into a set of physical directions. Color $SU(3)$ moves vertically through red, green, and blue; charged weak transformations move horizontally

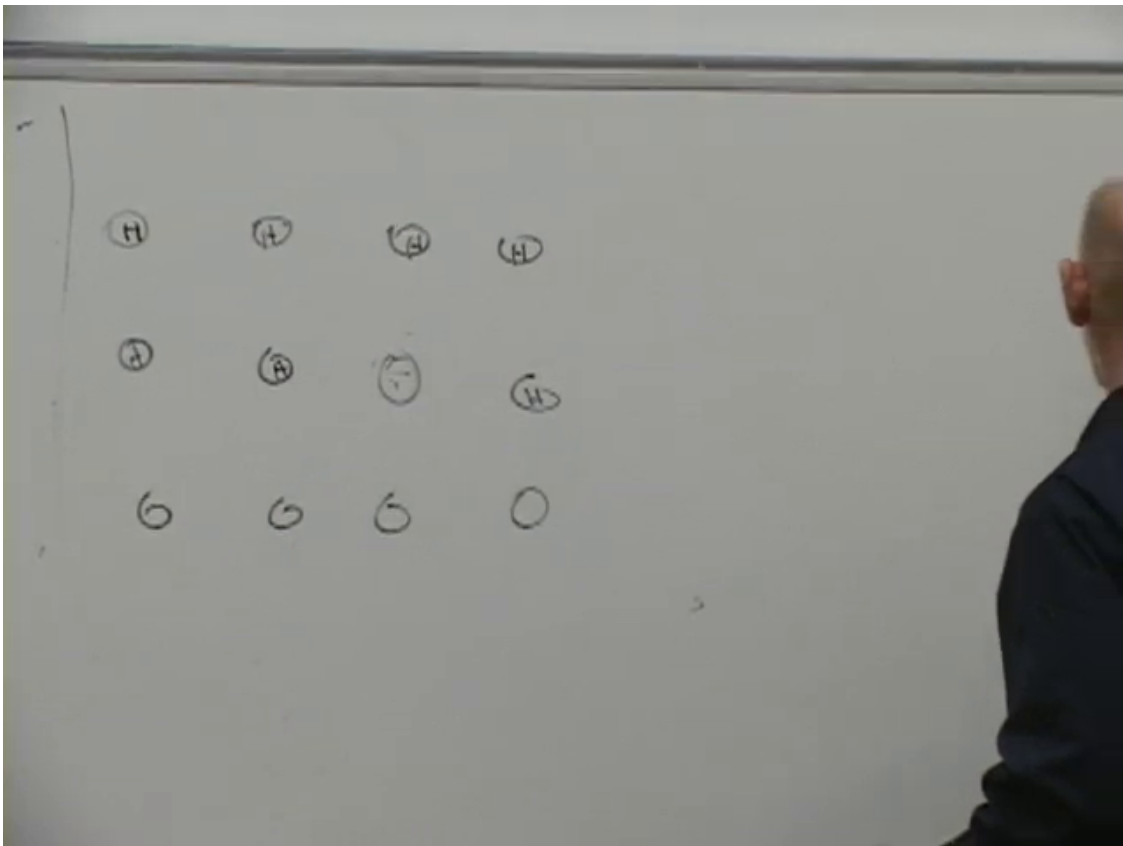


Figure 1.8: The heads–tails lattice divided by a dashed domain wall. Opposite boundary conditions force the system to contain an interface between the two degenerate ordered states.

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through quark and lepton doublets; a phase symmetry supplies the preliminary $U(1)$ factor. Gauge boson emission realizes these group transformations as interactions.

The weakness of beta decay is then traced not to an exceptionally tiny g_w , but to the heavy mediator. In distance space, mass cuts off the propagator beyond the Compton scale $\hbar/(m_W c)$. In momentum space,

$$\mathcal{A}_{\text{weak}} \sim \frac{g_w^2}{k^2 + m_W^2} \simeq \frac{g_w^2}{m_W^2} \quad (1.37)$$

at nuclear momenta. Old low-energy experiments measured this combination as the Fermi constant.

The extended classroom discussion clarifies what this formula does and does not mean. Internal W lines are virtual and off shell; energy–momentum is still conserved. A hard enough probe can produce a real W , but then it has created a different final-state process. Collision products are not a literal list of pre-existing constituents. Temperature changes the population of energetic projectiles, photons can interact through charged virtual particles, and Bose statistics is distinct from a dynamical force.

Finally, $SU(2)$ requires a missing neutral gauge field related to the Z , and the heads–tails lattice supplies the first model of spontaneous symmetry breaking. The next step is to combine those two unfinished threads: electroweak mixing and the Higgs mechanism.